Filter-matched rate calibration for variable-rate PING training

exp077 · 5 August 2026 · Draft

Abstract

Poisson rate coding provides a simple interface between static images and spiking neural networks, but the encoding rate does not by itself determine the signal available to a downstream circuit. Synaptic decay, conductance-dependent membrane integration, and finite observation windows reshape both the strength and variability of the representation. This makes input-rate selection especially important for sparse conductance-based PING networks, where low rate regimes that are computationally efficient may also erase class structure before learning begins. We developed a filter-matched calibration of MNIST inputs by propagating pixel spike trains through the target AMPA and subthreshold membrane dynamics, measuring the resulting empirical response distributions, and comparing them with a linear transfer-function model. The empirical response table preserved non-Gaussian finite-window effects that the stationary approximation did not capture accurately. Sampled feature images recovered recognizable digit structure as rate increased, whereas the lowest-rate images remained sparse and did not satisfy all image-level validation tolerances. These results established an empirical basis for selecting candidate input rates before decoder or recurrent-network training; classification thresholds remain to be measured.

Purpose and scope

The question is:

> At a given encoding rate, does the temporally filtered image still contain enough information for an ANN to recognize the digit?

The target PING architecture has 784 pixel channels feeding conductance-based excitatory cells. Its fixed synaptic and membrane equations define the features, without trained input, recurrent, or output weights. The ANN thus measures filtered-input decodability independently of a trained PING network.

The study was organized to:

1. Convert 0.25–25 Hz Poisson inputs into static, filter-matched features.
2. Measure their distributions and test a linear variance prediction.
3. Construct complete MNIST feature images from the empirical response table.
4. Keep the official MNIST test set held out for subsequent decoder evaluation.

- Nonlinear ANN above chance: this decoder can access class evidence.

- Nonlinear ANN at chance: this representation and decoder cannot extract reliable evidence.
- Nonlinear ANN succeeds but linear decoder fails: the evidence is not linearly accessible.

The completed analyses characterized the input representation rather than classification performance. Any later ANN threshold would be a decoder-relative decodability edge, not an absolute information boundary or a prediction of PING accuracy.

Methods

1. **Generated and validated filter-matched pixel features.** Every image used a 200 ms presentation and 0.1 ms timestep. The tested rates were 0.25, 0.5, 0.75, 1, 1.5, 2, 2.5, 3, 4, 5, 10, 25 Hz, with dense sampling below 5 Hz and an upper bound of 25 Hz, which has previously been show to have good performance in trained PING networks. Seeds 42, 43, 44 defined independent feature and empirical response table runs.

   The rate grid and deterministic MNIST training and validation indices were recorded before analysis; the official MNIST test set was held out and not loaded. Pixel intensities x lay in [0, 1]. If r was the encoding rate at pixel intensity one, the expected pixel rate was r x.

   Each pixel drove an independent AMPA synapse and uncoupled, non-spiking excitatory membrane, capturing AMPA decay, conductance-dependent integration, and within-window timing that raw Poisson encoded spike counts would omit. This probe has no threshold, reset, recurrence, or trained weights.

   The complete feature path is

   $$S_i(t) \to g_i^{\text{pix}}(t) \to v_i(t) \to z_i \to \text{ANN}. \quad (1)$$

   Here S_i, g^{pix}_i, v_i, and z_i are pixel i's spike train, AMPA conductance, subthreshold voltage, and time-averaged feature; t is time and ANN is the classifier.

   At each timestep, we drew the encoded input according to

   $$S_i(t) \sim \text{Bernoulli}(r\Delta t x_i). \quad (2)$$

   Here x_i is grayscale pixel-intensity, r is the encoding rate in spikes per second at pixel-intensity one, Δt is the timestep in seconds, and Bernoulli is an independent binary draw.

   We updated the AMPA conductance using

   $$g_i^{\text{pix}}(t) = \beta_{\text{AMPA}} g_i^{\text{pix}}(t-1) + w_{\text{probe}} S_i(t). \quad (3)$$

   $$\beta_{\text{AMPA}} = \exp\left(-\frac{\Delta t}{\tau_{\text{AMPA}}}\right). \quad (4)$$

   Here β_{AMPA} is one-timestep conductance retention, τ_{AMPA} is AMPA decay time, w_{probe} is conductance added per spike, and exp is the exponential function. Other symbols follow Equation 2. The primary 1.2 μS probe is the target architecture's nominal mean initial pixel-to-excitatory conductance; 0.6 and 2.4 μS test half and double scale for comparison.

   We integrated the membrane without thresholding or resetting:

$$C_E \frac{dv_i}{dt} = g_{\mathrm{L,E}}(E_L - v_i) + g_i^{\mathrm{pix}}(t)(E_e - v_i). \quad (5)$$

$$\tau_{\mathrm{eff}}, i(t) = \frac{C_E}{g_{\mathrm{L,E}} + g_i^{\mathrm{pix}}(t)}. \quad (6)$$

Here C_E is excitatory capacitance; $g_{L,E}$ is leak conductance; E_L and E_e are leak and AMPA reversal potentials; and $\tau_{eff,i}$ is the instantaneous conductance-dependent time constant. Other symbols follow Equation 1.

We initialized voltage at E_L and conductance at zero, simulated one complete presentation, and calculated

$$z_i = \frac{1}{T} \int_0^T (v_i(t) - E_L)\, \mathrm{d}t. \quad (7)$$

Here T is presentation duration and z_i is mean baseline-subtracted voltage. We did not divide by rate or disclose it to either decoder, so lower rates retain their weaker, noisier signal.

Focused tests covered per-timestep spike probability, count moments, pixel independence, AMPA decay, exponential-Euler voltage updates, deterministic replay, agreement with one uncoupled target-PING excitatory cell, and spike-timing sensitivity.

2. **Generated and evaluated the empirical response table.** Before generating the response table, we evaluated candidate draw counts K of 64, 128, 256, 512, 1,024, and 2,048 using a fixed convergence rule. The deterministic comparison covered 270 conditions: six intensities, five rates, all three probe conductances, and all three registered seeds. For each candidate, we compared the first K draws with the next K independent draws.

   Mean and unbiased-variance discrepancies were divided by locked absolute or relative tolerances, whichever was larger. Both metrics had to keep their 95th percentile at or below 1 and their maximum at or below 2. The smallest passing K selected the final draw count.

   This procedure selected K = 2048. We then ran that many independent single-pixel draws for every grayscale level, registered rate, probe conductance, and seed. The retained float32 array has shape 3 × 3 × 12 × 256 × 2048 in the ordered axes seed, probe conductance, rate, intensity, and draw. Its payload occupies 226492416 bytes in local scratch and is authenticated by the SHA-256 digest 5184788979b5fa7fa9a3f38936399b8e2724d63914019f1306a7171981b36783.

   The final response table estimated the conditional mean

$$\hat{\mu}_z(x, r, w_{\mathrm{probe}}) = \frac{1}{K} \sum_{k=1}^{K} z^k. \quad (8)$$

   and conditional variance

$$\hat{\sigma}_z^2(x, r, w_{\mathrm{probe}}) = \frac{1}{K-1} \sum_{k=1}^{K} \left(z^k - \hat{\mu}_z\right)^2. \quad (9)$$

Here x, r, and w_{probe} are pixel-intensity, rate, and probe conductance; $z^{(k)}$ is draw k; K is the draw count; and Equations 8 and 9 estimate the conditional mean and variance.

We retained all K simulated values as the empirical response table. Complete feature images drew one value according to

$$J \sim \mathrm{DiscreteUniform}(1, \cdots, K), \quad z_{\mathrm{sample}} = z^J. \quad (10)$$

Here J was uniform on the K draws and z_{sample} was the sampled feature; other symbols follow Equations 8 and 9. Mean and variance summarize noise, but the ANN samples the retained empirical values because low-rate responses may be zero-heavy, skewed, and non-Gaussian.

The generator wrote 96 deterministic, independently authenticated intensity chunks. Exact replay passed for one predeclared chunk from every seed. The complete grid, finite values, physical bounds, zero-intensity resting response, independent streams, independently recomputed moments, fresh direct simulations, and float32 fidelity all passed their checks. The low-rate representative condition had zero fraction 0.986, skewness 8.666, and 85 distinct retained values, preserving its discrete, non-Gaussian structure.

The required monotonicity check did not pass. It compared every adjacent intensity and rate pair using the predeclared 3-standard-error tolerance. 5 of 9180 intensity pairs exceeded it, while 0 of 8448 rate pairs did. The response table therefore did not meet the monotonicity validation criterion.

3. **Compared empirical variance with a linear-filter prediction.** We tested whether a local linear approximation explained the measured feature variance. This diagnostic was not used to select a training range.

   For mean spike rate λ, stationary mean conductance is

$$|g|_\lambda = \lambda w_{\mathrm{probe}} \tau_{\mathrm{AMPA}}. \quad (11)$$

   and stationary mean voltage is

$$|v|_\lambda = \frac{g_{\mathrm{L,E}} E_L + |g|_\lambda E_e}{g_{\mathrm{L,E}} + |g|_\lambda}. \quad (12)$$

   Here λ is encoding rate times pixel-intensity; bar(g)$_\lambda$ and bar(v)$_\lambda$ are stationary mean conductance and voltage. Other symbols follow Equations 3–7.

   Linearizing the synapse and membrane around that operating point gives

$$G_{\lambda(\omega)} = \frac{w_{\mathrm{probe}}}{i\omega + \frac{1}{\tau_{\mathrm{AMPA}}}} \cdot \frac{E_e - |v|_\lambda}{i\omega C_E + g_{\mathrm{L,E}} + |g|_\lambda}. \quad (13)$$

   Here G_λ (ω) is the local synapse-plus-membrane transfer function; ω is angular frequency; and i is the imaginary unit. Because mean conductance changes gain and effective membrane time constant, Equation 13 defines a family of responses.

   The finite averaging window contributes

$$A_{T(\omega)} = \frac{1 - \exp(-i\omega T)}{i\omega T}. \quad (14)$$

$$H_{\lambda(\omega)} = A_{T(\omega)} G_{\lambda(\omega)}. \quad (15)$$

Here $A_T(\omega)$ is the duration-T averaging response; $H_\lambda(\omega)$ is the complete response from input fluctuation to averaged feature; T is presentation duration; and exp is the exponential function.

For centred ideal Poisson input,

$$S_{\mathrm{in}(\omega)} = \lambda. \quad (16)$$

$$S_{z(\omega)} = \left|H_{\lambda(\omega)}\right|^2 S_{\mathrm{in}(\omega)}. \quad (17)$$

$$\mathrm{Var}_{\mathrm{linear}(z)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left|H_{\lambda(\omega)}\right|^2 S_{\mathrm{in}(\omega)} \, \mathrm{d}\omega. \quad (18)$$

Here $S_{in}(\omega)$ and $S_z(\omega)$ are input and predicted output power spectral densities; $|H_\lambda(\omega)|^2$ is transmitted noise power; $\mathrm{Var}_{linear}(z)$ is predicted feature variance; and π is the circle constant. Appendix A derives Equations 11–18.

We evaluated every distinct calibration point, compared predicted with empirical variance, and validated low-, middle-, and high-drive gains by sinusoidally modulating the numerical probe. This stationary, continuous-time, linearized Poisson model approximates the finite, discrete Bernoulli simulation and remains diagnostic only.

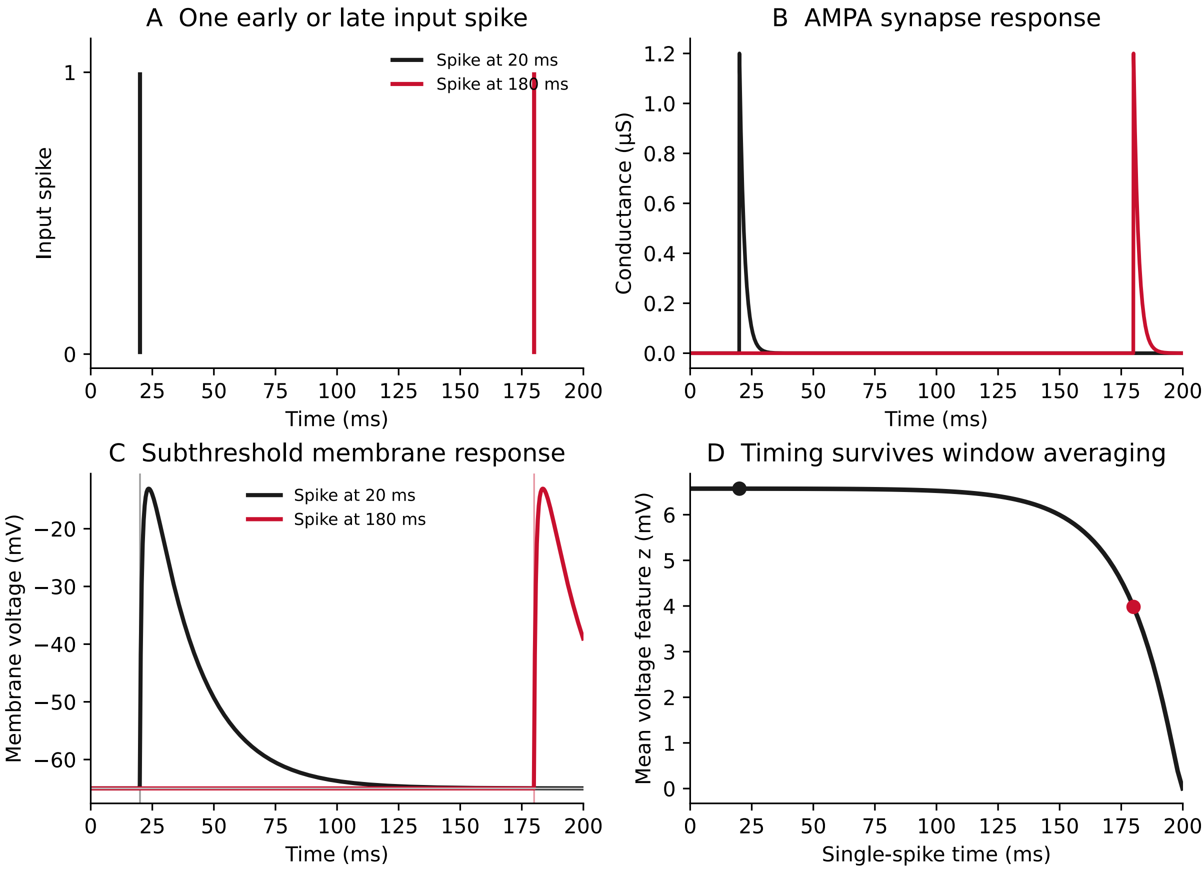
4. **Constructed and compared complete feature images.** We used only the official 60,000-image MNIST training partition. Indices 0–54,999 and 55,000–59,999 were reserved for decoder training and validation, respectively; the official MNIST test set was held out and not loaded. For each uint8 pixel, the sampler used its exact 0–255 intensity index and selected one of the $K = 2048$ authenticated empirical draws with independent, deterministic pixel and image streams.

   The direct comparison used training images 0–15, 28 × 28 pixels, eight independent replicates, all three conductances, and rates 0.25, 3, and 25 Hz. We compared pooled pixel moments, image-level moments, zero fractions, absolute and relative differences, and the spatial correlation of per-pixel means. The thresholds and streams were locked before outcomes. This validation was required before ANN training; its low-rate checks did not all pass, so no decoder followed.

Results

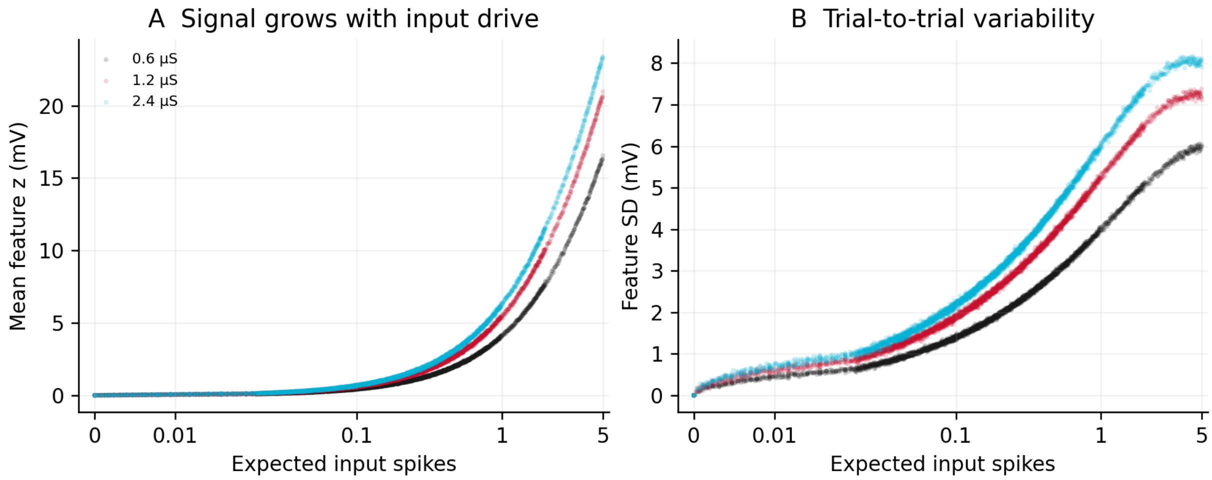
We characterized the filter-matched pixel response, empirical response table, linear approximation, and complete feature images. The response table did not meet its monotonicity criterion, and the complete feature images did not meet all low-rate image-level tolerances. Decoder training, psychometric evaluation, and a training-range decision were not performed and remain incomplete.

Filter-matched feature generation



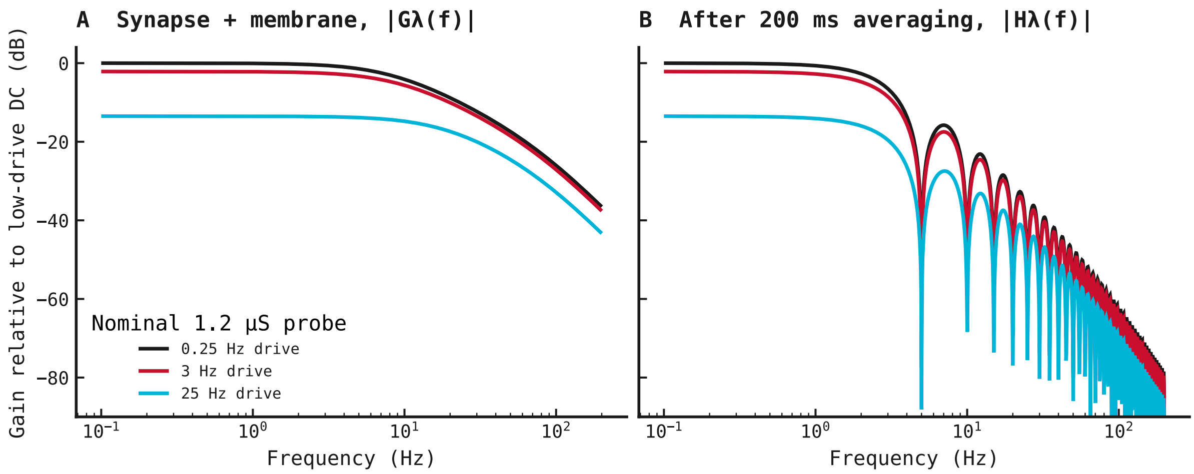
Finite-window timing sensitivity. Panel A places one input spike at 20 or 180 ms. Panels B and C pass those inputs through the registered AMPA synapse and subthreshold membrane. Panel D repeats the simulation across 101 single-spike times and plots the presentation-averaged feature z. Conductance and voltage rise after each spike and then relax through their respective decay dynamics; z falls for late spikes because the fixed 200 ms window truncates more of their response.

Empirical response table



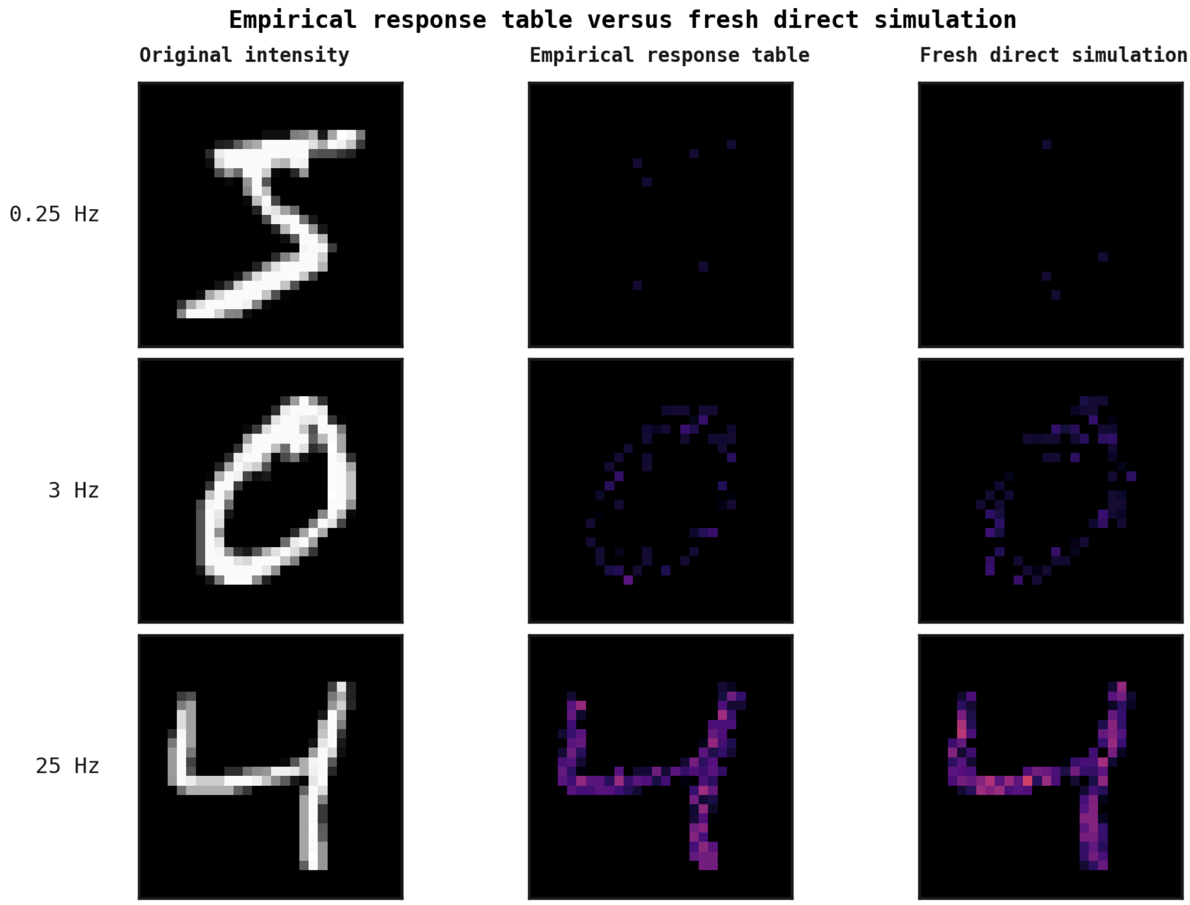
Signal and variability across the empirical response table. For every rate–intensity condition, Panel A plots the mean feature z and Panel B its standard deviation across the authenticated draws, against encoding rate × normalized pixel intensity × 200 ms. Each point is one observed rate–intensity condition, and colour encodes probe conductance. Both moments rise because larger expected spike counts deliver more stochastic conductance; larger probes produce larger voltage excursions.

Linear-filter prediction



Analytically calculated linearized frequency response at the nominal 1.2 µS probe. The plotted functions take modulation frequency, mean input spike rate, and probe conductance as inputs; no simulated traces are plotted. Frequency here is the frequency of a small sinusoidal modulation of the input spike rate, about mean operating rates of 0.25, 3, or 25 Hz. Panel A evaluates the synapse-plus-membrane transfer function on a 0.1–200 Hz frequency grid. Panel B multiplies it by the analytical 200 ms rectangular-window averaging response. Gain is referenced to the low-drive DC value. Mean conductance lowers and shifts the membrane response, the AMPA and membrane terms produce the smooth low-pass roll-off, and rectangular averaging introduces nulls at multiples of 5 Hz.

Complete feature images



Empirical response table samples versus fresh direct simulations. Rows show one original MNIST image, one authenticated empirical response table sample, and one fresh direct simulation at 0.25, 3, and 25 Hz for the nominal 1.2 µS probe, using common 0–65 mV limits. Low-rate images are sparse because most pixels receive no spike; increasing rate reveals the intensity pattern because spatial signal becomes larger relative to independent sampling noise.

Mixed-rate decoder training

Not yet performed.

ANN psychometric curve and thresholds

Not yet performed.

Training-range decision

Not yet performed.

Relation to prior work

Filtered conductance shot noise can violate fixed-time-constant models [**1**], and Gaussian approximations can miss skewed responses [**2**], supporting the empirical response table. Decoder performance is not absolute neural information [**3**], and linear and nonlinear decoders can differ [**4**], motivating a decoder-relative edge and both decoder diagnostics.

References

1. Wolff & Lindner: *Mean, Variance, and Autocorrelation of Subthreshold Potential Fluctuations Driven by Filtered Conductance Shot Noise.* Neural Computation, 2010. doi:10.1162/neco.2009.02-09-958
2. Brigham & Destexhe: *Nonstationary Filtered Shot-Noise Processes and Applications to Neuronal Membranes.* Physical Review E, 2015. doi:10.1103/PhysRevE.91.062102
3. Quian Quiroga & Panzeri: *Extracting Information from Neuronal Populations: Information Theory and Decoding Approaches.* Nature Reviews Neuroscience, 2009. doi:10.1038/nrn2578
4. Warland, Reinagel & Meister: *Decoding Visual Information From a Population of Retinal Ganglion Cells.* Journal of Neurophysiology, 1997. doi:10.1152/jn.1997.78.5.2336

Appendix A: linear-filter derivation

This appendix derives Equations 11–18 from the same AMPA synapse and subthreshold membrane used by the empirical probe. It replaces the finite Bernoulli drive with a continuous-time Poisson point process and linearizes the membrane around the mean drive at each calibration condition.

A.1 Stationary operating point

Represent pixel i's idealized spike train and synaptic conductance by

$$s_{i(t)} = \sum_k \delta(t - t_k), \quad \frac{dg_{i(t)}}{dt} = -\frac{g_{i(t)}}{\tau_{\text{AMPA}}} + w_{\text{probe}} s_{i(t)}. \quad \text{(A1)}$$

Here s_i (t) is a sum of Dirac impulses; t_k is spike k's time; δ is the Dirac delta; g_i (t) is AMPA conductance; τ_{AMPA} is its decay time; w_{probe} is conductance added per spike; and t is time. Equation A1 is the continuous-time counterpart of the decay-then-add update in Equation 3.

For stationary Poisson rate λ, the mean conductance satisfies

$$0 = -\frac{|g|_\lambda}{\tau_{\text{AMPA}}} + w_{\text{probe}}\lambda, \quad |g|_\lambda = \lambda w_{\text{probe}} \tau_{\text{AMPA}}. \quad \text{(A2)}$$

Here λ is encoding rate times pixel-intensity, and bar(g)$_\lambda$ is stationary mean conductance. The zero on the left states that the mean no longer changes with time.

Setting the mean membrane derivative in Equation 5 to zero gives

$$0 = g_{\text{L,E}}(E_L - |v|_\lambda) + |g|_\lambda(E_e - |v|_\lambda), \quad |v|_\lambda = \frac{g_{\text{L,E}}E_L + |g|_\lambda E_e}{g_{\text{L,E}} + |g|_\lambda}. \quad \text{(A3)}$$

Here bar(v)$_\lambda$ is stationary mean voltage; $g_{\text{L,E}}$ is leak conductance; and E_L and E_e are leak and AMPA reversal potentials. Other symbols follow Equation A2.

A.2 Local synapse-plus-membrane response

Write each signal as its operating point plus a small fluctuation:

$$g_i = |g|_\lambda + \delta g_i, \quad v_i = |v|_\lambda + \delta v_i, \quad s_i = \lambda + \delta s_i. \quad \text{(A4)}$$

Here δg_i, δv_i, and δs_i are complete conductance, voltage, and input perturbation variables around the means defined in Equations A2 and A3.

To derive Equation A5, substitute the conductance and input decompositions from Equation A4 into the synapse equation, Equation A1:

$$\frac{d(|g|_\lambda + \delta g_i)}{dt} = -\frac{|g|_\lambda + \delta g_i}{\tau_{\text{AMPA}}} + w_{\text{probe}}(\lambda + \delta s_i).$$

The operating point is stationary, so its time derivative is zero. Expanding the right-hand side then gives

$$\frac{d\delta g_i}{dt} = \left(-\frac{|g|_\lambda}{\tau_{\text{AMPA}}} + w_{\text{probe}}\lambda\right) - \frac{\delta g_i}{\tau_{\text{AMPA}}} + w_{\text{probe}}\delta s_i.$$

The parenthesized stationary terms cancel by Equation A2. The remaining perturbation dynamics are

$$\frac{d\delta g_i}{dt} = -\frac{\delta g_i}{\tau_{\text{AMPA}}} + w_{\text{probe}}\delta s_i. \quad \text{(A5)}$$

To derive Equation A6, substitute the conductance and voltage decompositions from Equation A4 into the membrane equation, Equation 5:

$$C_E\frac{d(|v|_\lambda + \delta v_i)}{dt} = g_{\text{L,E}}(E_L - |v|_\lambda - \delta v_i) + (|g|_\lambda + \delta g_i)(E_e - |v|_\lambda - \delta v_i).$$

As above, the stationary mean has zero time derivative. Expanding and grouping the right-hand side gives

$$C_E\frac{d\delta v_i}{dt} = \big(g_{\text{L,E}}(E_L - |v|_\lambda) + |g|_\lambda(E_e - |v|_\lambda)\big) - \big(g_{\text{L,E}} + |g|_\lambda\big)\delta v_i + (E_e - |v|_\lambda)\delta g_i - \delta g_i \delta v_i.$$

The first parenthesized group is zero by Equation A3. The final product is second order in the small perturbations and is omitted by the local linear approximation. This leaves

$$C_E\frac{d\delta v_i}{dt} = -\big(g_{\text{L,E}} + |g|_\lambda\big)\delta v_i + (E_e - |v|_\lambda)\delta g_i. \quad \text{(A6)}$$

Here C_E is excitatory-cell capacitance. All other symbols follow Equations A1–A4.

To derive Equation A7, take the Fourier transform of Equation A5. The derivative becomes $i\omega$ times the transformed signal:

$$i\omega\delta g_{i(\omega)} = -\frac{\delta g_{i(\omega)}}{\tau_{\text{AMPA}}} + w_{\text{probe}}\delta s_{i(\omega)}.$$

Move both conductance terms to the left, then divide by their common factor:

$$\left(i\omega + \frac{1}{\tau_{\text{AMPA}}}\right)\delta g_{i(\omega)} = w_{\text{probe}}\delta s_{i(\omega)},$$

$$\delta g_{i(\omega)} = \frac{w_{\text{probe}}}{i\omega + \frac{1}{\tau_{\text{AMPA}}}}\delta s_{i(\omega)}. \quad \text{(A7)}$$

Equation A8 follows in the same way from Equation A6. Fourier transformation first gives

$$i\omega C_E \delta v_{i(\omega)} = -\big(g_{\text{L,E}} + |g|_\lambda\big)\delta v_{i(\omega)} + (E_e - |v|_\lambda)\delta g_{i(\omega)}.$$

Collect the voltage terms on the left:

$$\left(i\omega C_E + g_{\mathrm{L,E}} + |g|_\lambda\right)\delta v_{i(\omega)} = (E_e - |v|_\lambda)\delta g_{i(\omega)}.$$

Dividing by the coefficient of δvᵢ (ω) gives

$$\delta v_{i(\omega)} = \frac{E_e - |v|_\lambda}{i\omega C_E + g_{\mathrm{L,E}} + |g|_\lambda}\delta g_{i(\omega)}. \quad \text{(A8)}$$

Here ω is angular frequency, i is the imaginary unit, and each argument ω denotes a Fourier-domain signal.

Equations A5–A8 form a **linear time-invariant (LTI)** system after linearization about the operating point λ. Because the system has been linearized about the operating point, it is locally linear and time-invariant, so the standard transfer-function formalism applies. The canonical frequency-domain input–output relationship is

$$\delta v_{i(\omega)} = G_{\lambda(\omega)}\delta s_{i(\omega)}.$$

Here G_λ (ω) is the local transfer function relating input spike-train perturbations to membrane-voltage perturbations. For non-zero input, the input–output relationship can be rearranged as

$$G_{\lambda(\omega)} = \frac{\delta v_{i(\omega)}}{\delta s_{i(\omega)}}.$$

Thus the ratio follows from the standard input–output relationship rather than serving as an independent definition. To derive G_λ (ω), start from Equation A7:

$$\delta g_{i(\omega)} = \frac{w_{\mathrm{probe}}}{i\omega + \frac{1}{\tau_{\mathrm{AMPA}}}}\delta s_{i(\omega)}.$$

Substitute this expression for δg_i (ω) into Equation A8:

$$\delta v_{i(\omega)} = \frac{E_e - |v|_\lambda}{i\omega C_E + g_{\mathrm{L,E}} + |g|_\lambda}\left(\frac{w_{\mathrm{probe}}}{i\omega + \frac{1}{\tau_{\mathrm{AMPA}}}}\delta s_{i(\omega)}\right).$$

Reorder the scalar factors and factor out δs_i (ω):

$$\delta v_{i(\omega)} = \left(\frac{w_{\mathrm{probe}}}{i\omega + \frac{1}{\tau_{\mathrm{AMPA}}}} \cdot \frac{E_e - |v|_\lambda}{i\omega C_E + g_{\mathrm{L,E}} + |g|_\lambda}\right)\delta s_{i(\omega)}.$$

Comparing this result with the canonical LTI relationship identifies the coefficient multiplying δs_i (ω) as the transfer function:

$$G_{\lambda(\omega)} = \frac{w_{\mathrm{probe}}}{i\omega + \frac{1}{\tau_{\mathrm{AMPA}}}} \cdot \frac{E_e - |v|_\lambda}{i\omega C_E + g_{\mathrm{L,E}} + |g|_\lambda}. \quad \text{(A9)}$$

Equation A9 is therefore the synaptic filter multiplied by the membrane filter, not an arbitrarily chosen ratio. Its λ-dependence captures the change in membrane gain and effective time constant with mean conductance.

A.3 Finite-window averaging

Start from the baseline-subtracted Step 1 feature in Equation 7:

$$z_i = \frac{1}{T}\int_0^T \left(v_{i(t)} - E_L\right) \mathrm{d}t.$$

Within the stationary linearized model used in this appendix, substitute the voltage decomposition from Equation A4:

$$z_i = \frac{1}{T}\int_0^T \left(|v|_\lambda + \delta v_{i(t)} - E_L\right) \mathrm{d}t.$$

The operating-point voltage and leak reversal potential are constant in time, so their integral is their difference multiplied by T. Dividing by T gives

$$z_i = |v|_\lambda - E_L + \frac{1}{T}\int_0^T \delta v_{i(t)} \, \mathrm{d}t.$$

Define the operating-point feature and its perturbation by

$$|z|_\lambda = |v|_\lambda - E_L, \quad \delta z_i = z_i - |z|_\lambda.$$

Subtracting the operating-point feature from the expanded expression for z_i cancels the constant term and leaves

$$\delta z_i = \frac{1}{T}\int_0^T \delta v_{i(t)} \, \mathrm{d}t. \quad \text{(A10)}$$

Thus δz_i is the fluctuation around the operating-point feature, obtained by averaging the voltage perturbation over the 200 ms presentation duration T. To derive the frequency response, represent the averaging operation as a causal rectangular kernel a_T (u), where

$$a_{T(u)} = \frac{1}{T}, \quad 0 \le u \le T,$$

and a_T (u) is zero outside that interval. Applying this kernel as a moving-average filter gives

$$\delta z_{i(t)} = \int_{-\infty}^{\infty} a_{T(u)} \delta v_{i(t-u)} \, \mathrm{d}u = \frac{1}{T}\int_0^T \delta v_{i(t-u)} \, \mathrm{d}u.$$

At the end of the presentation, set t equal to T. To put the resulting integral into the same form as Equation A10, introduce the new variable

$$t' = T - u, \quad \mathrm{d}t' = -\,\mathrm{d}u.$$

When u is zero, t' is T; when u is T, t' is zero. The substitution therefore reverses the integration limits. Its minus sign reverses them back:

$$\delta z_{i(T)} = \frac{1}{T}\int_0^T \delta v_{i(T-u)} \, \mathrm{d}u = -\frac{1}{T}\int_T^0 \delta v_{i(t')} \, \mathrm{d}t' = \frac{1}{T}\int_0^T \delta v_{i(t')} \, \mathrm{d}t'.$$

The name of an integration variable has no effect on the value, so replacing t' with t recovers Equation A10 exactly.

For any time-domain function f(u), use the Fourier-transform convention

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(u) \exp(-i\omega u) \, \mathrm{d}u.$$

Here hat(f) (ω) is the Fourier transform of f(u). Applying this definition with f(u) equal to the rectangular kernel a_T (u), and naming the result A_T (ω), gives

$$A_{T(\omega)} = \int_{-\infty}^{\infty} a_{T(u)} \exp(-i\omega u)\,\mathrm{d}u = \frac{1}{T}\int_0^T \exp(-i\omega u)\,\mathrm{d}u.$$

Evaluating the integral and rearranging gives

$$A_{T(\omega)} = \frac{\exp(-i\omega T) - 1}{-i\omega T} = \frac{1 - \exp(-i\omega T)}{i\omega T}.$$

At zero frequency, the continuous limit of A_T (ω) is one, as expected: averaging does not change a constant signal. By the convolution theorem, the averaging filter multiplies the voltage spectrum:

$$\delta z_{i(\omega)} = A_{T(\omega)} \delta v_{i(\omega)}.$$

Substitute the input–voltage relation from Equation A9:

$$\delta z_{i(\omega)} = A_{T(\omega)} G_{\lambda(\omega)} \delta s_{i(\omega)}.$$

Define the complete local input–feature response by the canonical LTI relationship

$$\delta z_{i(\omega)} = H_{\lambda(\omega)} \delta s_{i(\omega)}.$$

Comparing the coefficients multiplying δs_i (ω) identifies H_λ (ω). The two components of Equation A11 are therefore

$$A_{T(\omega)} = \frac{1 - \exp(-i\omega T)}{i\omega T}, \quad H_{\lambda(\omega)} = A_{T(\omega)} G_{\lambda(\omega)}. \quad \text{(A11)}$$

Here A_T (ω) is the averaging-window response, exp is the exponential function, and H_λ (ω) is the complete response from input spike-train perturbation to averaged-voltage perturbation.

For the Bode comparison, the zero-frequency-normalized magnitude of the complete input–feature response is

$$B_{H(\omega)} = 20 \log_{10}\left(\frac{\left|H_{\lambda(\omega)}\right|}{\left|H_{\lambda(0)}\right|}\right). \quad \text{(A12)}$$

Here B_H is magnitude in decibels; | · | is complex magnitude; and $\log_{10}$ is the base-ten logarithm. Only H_λ is plotted because it includes the averaging window and is the response used to predict the ANN feature variance.

A.4 Poisson-noise variance

A centred, unit-amplitude Poisson point process has a flat two-sided input power spectral density:

$$S_{\mathrm{in}(\omega)} = \lambda. \quad \text{(A13)}$$

Here S_{in} (ω) is input power per unit angular frequency and λ is mean spike rate. Passing this noise through Equation A11 gives

$$S_{z(\omega)} = \left|H_{\lambda(\omega)}\right|^2 S_{\mathrm{in}(\omega)}. \quad \text{(A14)}$$

Here $S_z(\omega)$ is predicted feature-noise power and $|H_\lambda(\omega)|^2$ is the transmitted fraction of input-noise power. Under the two-sided angular-frequency convention, total predicted variance is

$$\mathrm{Var}_{\mathrm{linear}(z)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| H_{\lambda(\omega)} \right|^2 S_{\mathrm{in}(\omega)} \, \mathrm{d}\omega. \quad \text{(A15)}$$

Here $\mathrm{Var}_{\mathrm{linear}}(z)$ is predicted feature variance and π is the circle constant; the remaining symbols follow Equations A11–A14. This result is exact for the stated linear stationary model, but only approximate for the finite, discrete, conductance-dependent probe. Step 3 therefore compares it with empirical variance rather than assuming agreement.